



“India’s transition resembles a 4x400-metre relay race. The first leg was assembly. The second is component localisation. The third is design and the fourth will be the emergence of global Indian champions.”

VINOD SHARMA

CHAIRMAN, CII'S NATIONAL COMMITTEE ON ELECTRONICS MANUFACTURING



“The industry has evolved from basic assembly of imported parts to higher local value addition. Local capabilities have improved in PCB fabrication, testing, calibration.”

ASHOK GUPTA

DIRECTOR, OPTIEMUS INFRACOM

are still globally concentrated industries. Even where India is adding value, it is often doing so in lower-value sections such as enclosures, batteries, mechanics and sub-assembly.

The second challenge is capital. The founder of an EMS company who wished not to be named says “access to finance is a major constraint, especially for smaller manufacturers. Expanding a factory,

funding inventory or even financing exports can be difficult. For instance, for export to the US, someone must fund the passage, as the product will be at sea for around 60 days. The customer doesn't want to fund it as they are used to China funding it. The Indian supplier can't fund it, leading to critical gaps in the process.”

The third is fragmentation. Germany and China have clusters like Bavaria and Shenzhen, respectively, where suppliers, tooling shops, logistics providers and assemblers work within a short radius, creating powerful productivity gains. India's ecosystem is far more dispersed. That raises coordination costs and slows discovery. It also means Tier II and Tier III suppliers—often the backbone of component localisation—need more support in technology adoption and process upgrades. The fourth is culture and capability. Indian manufacturing still underuses software and technology. MSMEs (key component of the ecosystem) need digital tools to become globally competitive. Building a manufacturing powerhouse is not simply a matter of incentives and plants; it requires new ways of engineering, planning, process discipline and investment in research and development.

“The government is pushing a lot, but entrepreneurs also must pick up, spend more on R&D so that complete products can be made in India. Otherwise, India will primarily be a contract manufacturer to the world,” says Saurabh Agarwal, tax partner, EY India.

Previously, there were almost 100 local mobile handset brands, including Lemon, Karbonn Mobiles, Micromax, Intex and others. But almost all of them either folded up, scaled down operations or shifted gears to manufacturing only. The entry of low cost Chinese brands with new features and new models impacted their market share.

It took China decades to turn contract manufacturing into a broader industrial ecosystem. India has made a good start. Can it cross the finishing line? **BT**